

class 18

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Background

Pertussis (a.k.a. whooping cough) is a common lung infection caused by the bacteria B. Pertussis.

This can infect all ages but is most severe for those under 1 year of age.

The CDC track the number of [reported cases in the US](#): We can “scrapethis data with the **datapasta** package.

```
cdc <- data.frame(  
  year = c(1922L,1923L,1924L,1925L,  
           1926L,1927L,1928L,1929L,1930L,1931L,  
           1932L,1933L,1934L,1935L,1936L,  
           1937L,1938L,1939L,1940L,1941L,1942L,  
           1943L,1944L,1945L,1946L,1947L,  
           1948L,1949L,1950L,1951L,1952L,  
           1953L,1954L,1955L,1956L,1957L,1958L,  
           1959L,1960L,1961L,1962L,1963L,  
           1964L,1965L,1966L,1967L,1968L,1969L,  
           1970L,1971L,1972L,1973L,1974L,  
           1975L,1976L,1977L,1978L,1979L,1980L,  
           1981L,1982L,1983L,1984L,1985L,  
           1986L,1987L,1988L,1989L,1990L,  
           1991L,1992L,1993L,1994L,1995L,1996L,  
           1997L,1998L,1999L,2000L,2001L,
```

```

2002L,2003L,2004L,2005L,2006L,2007L,
2008L,2009L,2010L,2011L,2012L,
2013L,2014L,2015L,2016L,2017L,2018L,
2019L,2020L,2021L,2022L,2023L,2024L,2025L),
cases = c(107473,164191,165418,152003,
202210,181411,161799,197371,
166914,172559,215343,179135,265269,
180518,147237,214652,227319,103188,
183866,222202,191383,191890,109873,
133792,109860,156517,74715,69479,
120718,68687,45030,37129,60886,
62786,31732,28295,32148,40005,
14809,11468,17749,17135,13005,6799,
7717,9718,4810,3285,4249,3036,
3287,1759,2402,1738,1010,2177,2063,
1623,1730,1248,1895,2463,2276,
3589,4195,2823,3450,4157,4570,
2719,4083,6586,4617,5137,7796,6564,
7405,7298,7867,7580,9771,11647,
25827,25616,15632,10454,13278,
16858,27550,18719,48277,28639,32971,
20762,17972,18975,15609,18617,
6124,2116,3044,7063,22538,21996)
)

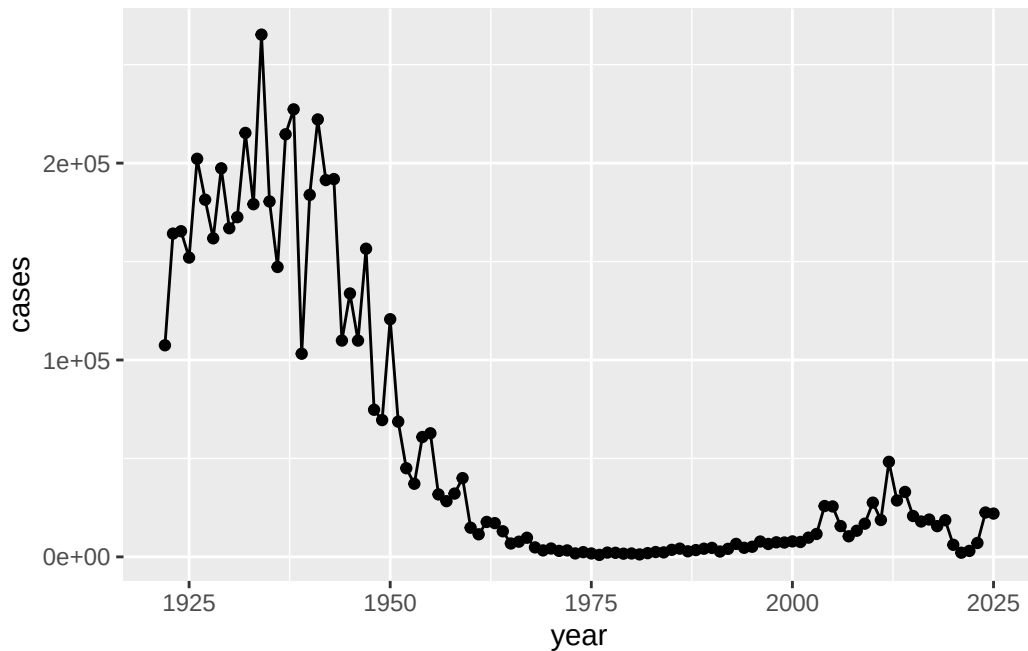
```

Q. Make a plot of year vs cases

```

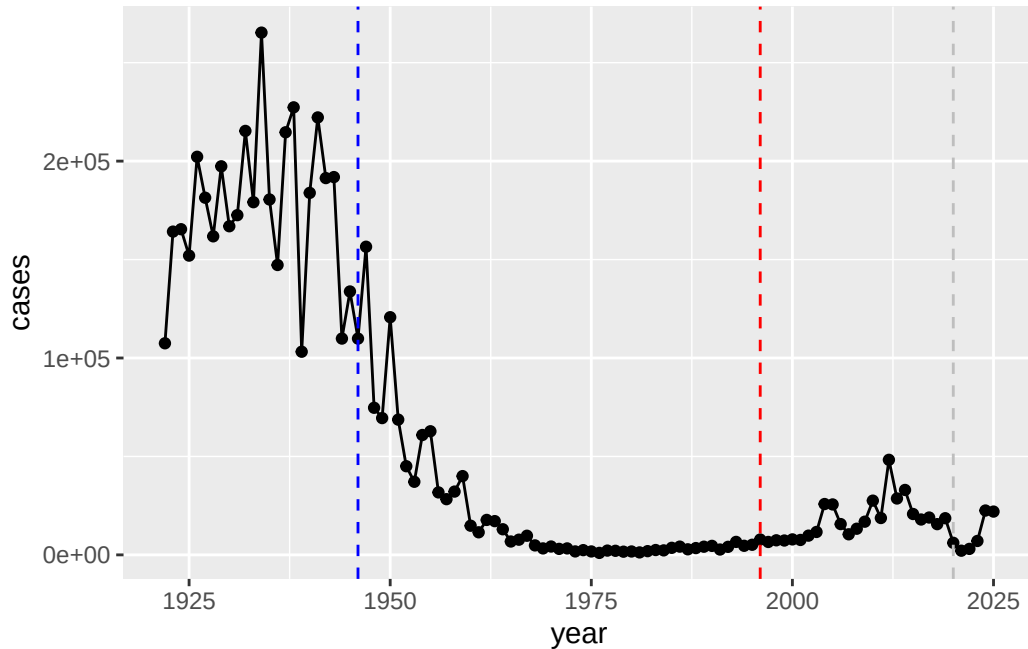
library(ggplot2)
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()

```



Q. Add some major milestones including the first wP vaccine roll out (1946), the switch to the newer aP vaccine (1996), the COVID years (2020).

```
library(ggplot2)
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col="blue", lty=2)+
  geom_vline(xintercept = 1996, col="red", lty=2)+
  geom_vline(xintercept = 2020, col="grey", lty=2)
```



Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

After 1946, the cases are low and after the switch the case numbers go up.

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

There were high case numbers because the AP induced protection wanes faster than wP.

Why is this vaccine-preventable disease on the upswing To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

#CMI-PB project [Computational Models of Immunity - Pertussis Boost](https://www.cmi-pb.org/) project aims to provide the scientific community with this very information.

They make their data available via JSON format returning API. We can read this in R with the `read_json()` function from the **jsonlite** package:

```
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. How many “wP” and “aP” individuals are in this subject table?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. What is the biological sex breakdown?

```
table(subject$biological_sex)
```

```
Female   Male
  112    60
```

Q. In terms of race and gender is this dataset representative of the US population?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

Let's read some more database tables:

```
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost	
1	1	1	-3	
2	2	1	1	
3	3	1	3	
4	4	1	7	
5	5	1	11	
6	6	1	32	

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

```
head(ab_titer)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

To analyze this data we need to first “join” (merge/link) the different tables so we have all the data in one place not spread across different tables.

We can use the `*_join()` family of functions from **dplyr** to do this

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
meta <- inner_join(subject, specimen)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White
4	1	wP	Female	Not Hispanic or Latino	White
5	1	wP	Female	Not Hispanic or Latino	White
6	1	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1
2	1986-01-01	2016-09-12	2020_dataset	2
3	1986-01-01	2016-09-12	2020_dataset	3
4	1986-01-01	2016-09-12	2020_dataset	4
5	1986-01-01	2016-09-12	2020_dataset	5
6	1986-01-01	2016-09-12	2020_dataset	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1		-3	0 Blood
2		1	1 Blood
3		3	3 Blood
4		7	7 Blood

5		11		14	Blood
6		32		30	Blood
	visit				
1	1				
2	2				
3	3				
4	4				
5	5				
6	6				

```
abdata <- inner_join(ab_titer, meta)
```

Joining with `by = join\_by(specimen\_id)`

```
head(abdata)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	UG/ML	2.096133	1	wP	Female
2	IU/ML	29.170000	1	wP	Female
3	IU/ML	0.530000	1	wP	Female
4	IU/ML	6.205949	1	wP	Female
5	IU/ML	4.679535	1	wP	Female
6	IU/ML	2.816431	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3		Blood
2	-3		Blood
3	-3		Blood

4		-3	0	Blood
5		-3	0	Blood
6		-3	0	Blood

	visit
1	1
2	1
3	1
4	1
5	1
6	1

Q. What Antibody isotypes are measured for these patients?

```
table(abdata$isotype)
```

IgE	IgG	IgG1	IgG2	IgG3	IgG4
6698	7265	11993	12000	12000	12000

Q. What antigens are reported?

```
table(abdata$antigen)
```

ACT	BETV1	DT	FELD1	FHA	FIM2/3	LOLP1	LOS	Measles	OVA
1970	1970	6318	1970	6712	6318	1970	1970	1970	6318
PD1	PRN	PT	PTM	Total	TT				
1970	6712	6712	1970	788	6318				

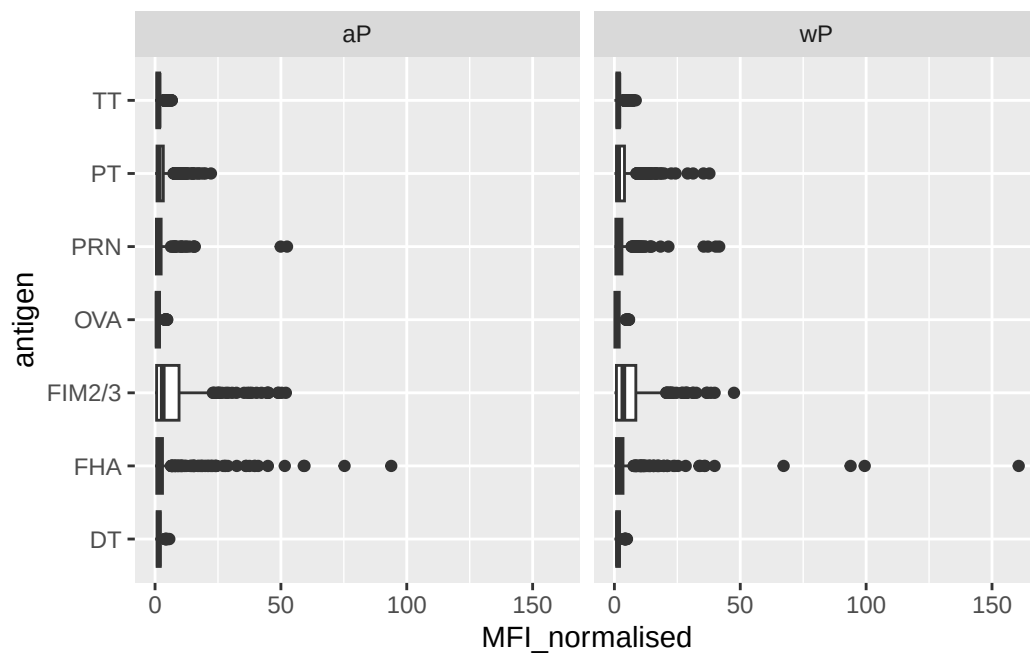
Let's focus on the IgG antigen and make a plot of MFI_normalised for all antigens.

```
igg <- abdata |>
  filter(isotype=="IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635

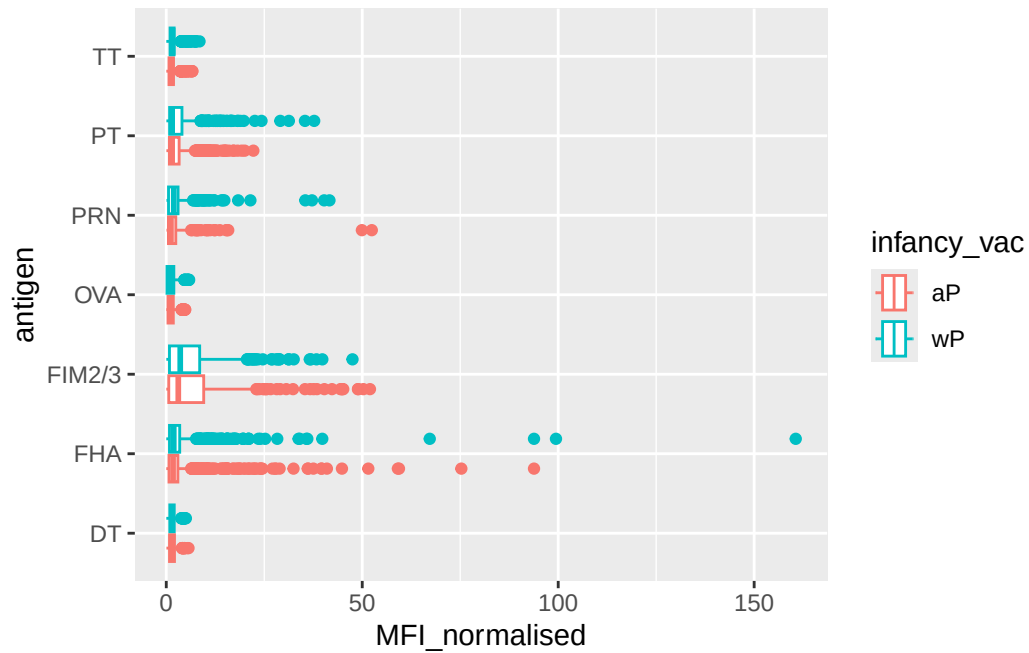
6	19	IgG	TRUE	FHA	60.76626	1.096457
	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex	
1	IU/ML	0.530000	1	wP	Female	
2	IU/ML	6.205949	1	wP	Female	
3	IU/ML	4.679535	1	wP	Female	
4	IU/ML	0.530000	3	wP	Female	
5	IU/ML	6.205949	3	wP	Female	
6	IU/ML	4.679535	3	wP	Female	
	ethnicity	race	year_of_birth	date_of_boost	dataset	
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type			
1	-3		0	Blood		
2	-3		0	Blood		
3	-3		0	Blood		
4	-3		0	Blood		
5	-3		0	Blood		
6	-3		0	Blood		
	visit					
1	1					
2	1					
3	1					
4	1					
5	1					
6	1					

```
ggplot(igg) +
  aes(MFI_normalised, antigen)+
  geom_boxplot()+
  facet_wrap(~infancy_vac)
```



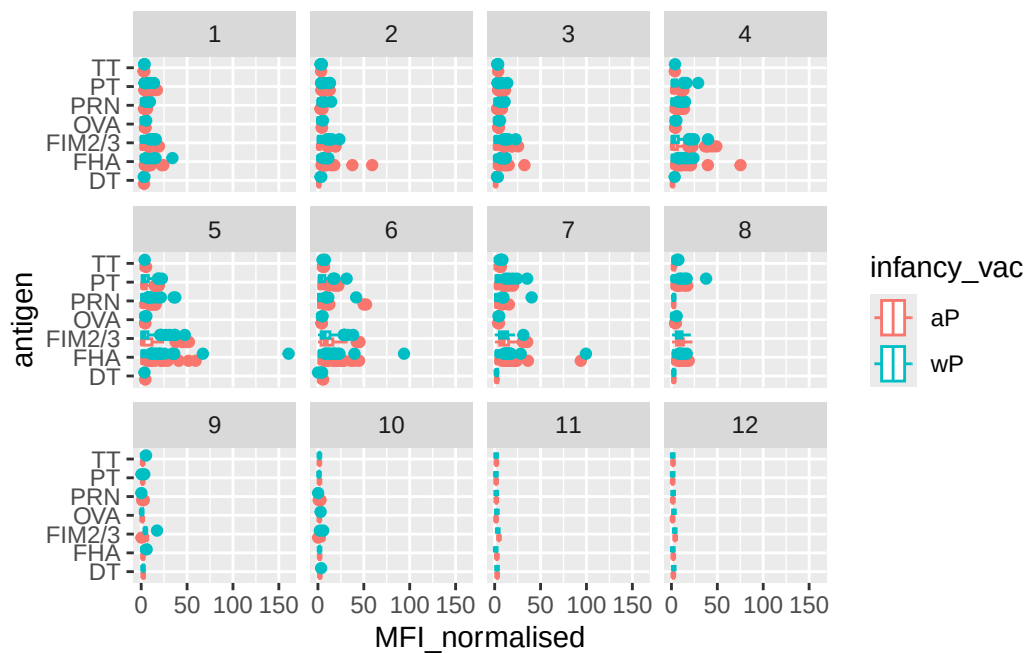
Q. Is there a difference for aP vs wP individuals with these values?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()
```



Q. Is there a temporal response - i.e. do values increase or decrease over time?

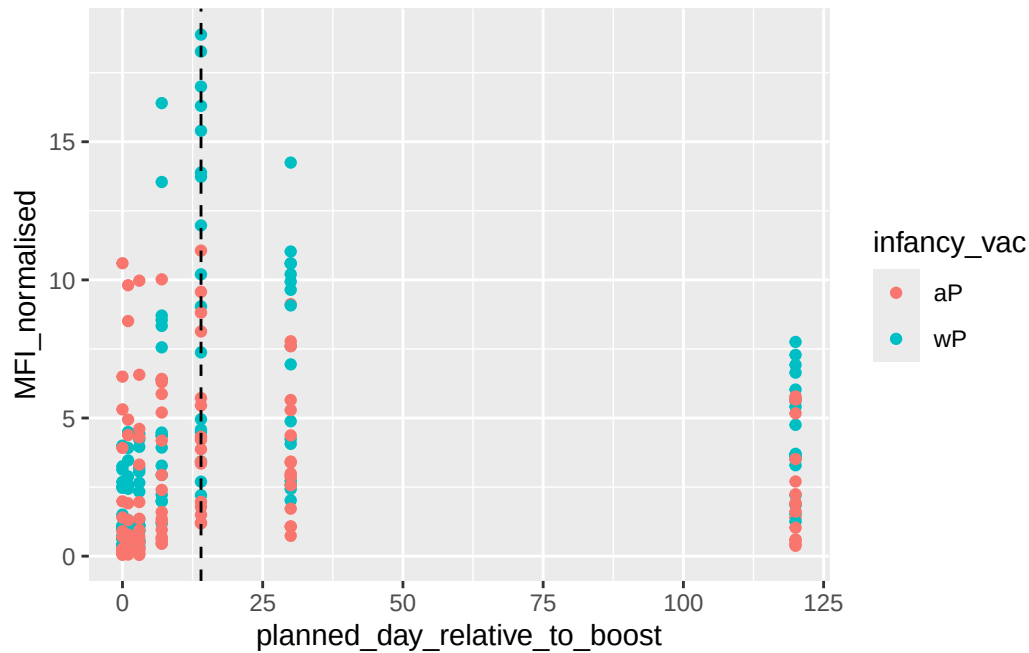
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()+
  facet_wrap(~visit)
```



Focus on “PT” Pertussis Toxin antigen

```
pt.igg.21 <- igg |> filter(antigen == "PT",
  dataset == "2021_dataset")
```

```
ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost, MFI_normalised,
    col=infancy_vac,
    group=subject_id)+
  geom_point()+
  geom_vline(xintercept =14, lty=2)
```



```
ggplot(pt.igg.21,
  aes(x = planned_day_relative_to_boost,
    y = MFI_normalised,
    col = infancy_vac)) +
  geom_point(aes(group = subject_id)) +
  geom_line(aes(group = subject_id)) +
  geom_smooth(se = FALSE, linewidth = 1.5) +
  geom_vline(xintercept = 0, lty = 2) +
  geom_vline(xintercept = 14, lty = 2) +
  labs(
    title = "CMI-PB 2021 dataset IgG PT",
    subtitle = "Dashed lines at day 0 (pre boost) and day 14 (post boost)",
    x = "Day relative to boost",
    y = "Normalised MFI"
  )
)
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

CMI-PB 2021 dataset IgG PT

Dashed lines at day 0 (pre boost) and day 14 (post boost)

